

Mark Zhou

Madison, WI | markjzouh@icloud.com | markjzouh.com | [linkedin.com/in/xinyun-mark-zhou](https://www.linkedin.com/in/xinyun-mark-zhou) | github.com/markjzouh

EDUCATION

University of Wisconsin–Madison

B.S. Computer Sciences, College of Letters & Science

Madison, WI

Sep. 2026 – May 2030 (expected)

George School

High School Diploma, College/University Preparatory Program

Newtown, PA

Sep. 2021 – May 2026

RESEARCH

Research Collaborator, Gerych Lab

Worcester Polytechnic Institute

Representation Learning & Multimodal ML (Remote); Advisor: Prof. Walter Gerych

Apr. 2026 – Present

- Leading implementation and experimental design for ConceptBasis, a manuscript in preparation on structured visual-attribute directions in vision-language embedding spaces
- Built the end-to-end research stack: curated concept dictionary, VLM-assisted exhaustive labeling, class-disjoint dataset splits, adapter training, sealed evaluation protocol, reproducible experiment records, and an interactive playground
- Combined joint regularized concept-to-embedding regression, concept-direction orthogonality, and image-text contrastive training to separate statistical confounding from geometric interference while preserving ordinary retrieval
- Designed evaluation across concept prediction, direction overlap, and class-disjoint compositional retrieval, including direct compositional supervision as a task-specific baseline

PROJECTS

RotateMate – Image Rotation ML Model

github.com/markjzouh/RotateMate-model

Personal Project

2025 – Present

- Built an end-to-end iOS ML system for detecting and correcting mis-rotated images, spanning custom data, PyTorch training, quantization, hardware-accelerated preprocessing, and Swift deployment

Tiaoqi Master – Chinese Checkers iOS App

github.com/markjzouh/tiaoqi-master

Swift Student Challenge 2026 Winner

2025 – 2026

- Built a native SwiftUI learning experience with an adaptive opponent, iterative-deepening minimax engine, contextual commentary, and procedural audio; selected as a 2026 Apple Swift Student Challenge winner

Competitive Programming Library

github.com/markjzouh/cp-library

Open Source Project

2023 – Present

- Maintained reusable C++ implementations of advanced data structures, graph algorithms, number theory, and geometry used in USACO and Codeforces competition

TECHNICAL SKILLS

Languages: Python, C++, Swift/SwiftUI, C, Java, HTML/CSS/JavaScript

Machine Learning: PyTorch, NumPy, scikit-learn, vision-language encoders, adapter training, multimodal labeling and evaluation

Systems & Tools: Linux, Git, GPU inference profiling, experiment orchestration, Xcode, iOS development

HONORS & LEADERSHIP

Honors: USACO Platinum; Codeforces Master; Apple Swift Student Challenge Winner; first-place finishes at CMIMC, PClassic Advanced, MACSO, and TJIOI

Leadership: Algorithms Club President & Co-Founder; Mathematics Tutor, George School